

Amazon Web Service

What is Amazon Web Service?

AWS is a secure cloud service platform that offers compute power, database, storage, content delivery, and other resources to help the business scale and grow.

- AWS was officially launched in 2006.
- AWS has over 200 fully featured services for a wide range of technologies, industries, and use cases.
- Client security is a significant concern for AWS. Their infrastructure is successfully satisfying the security concerns of the military, global banks, and other high-sensitivity organizations

Amazon Web Services (AWS) is a platform of web services that offers solutions for computing, storing, and networking, at different layers of abstraction. For example, you can attach volumes to a virtual machine - low level of abstraction - or store and retrieve data via a REST API – high level of abstraction. Use the services provided by AWS to host websites, run enterprise applications, and mine tremendous amounts of data. Web services are accessible via the internet by using typical web protocols (such as HTTP) and used by machines or by humans through a UI. The most prominent services provided by AWS are EC2, which offers virtual machines, and S3, which offers storage capacity. Services on AWS work well together: you can use them to replicate your existing local network setup, or you can design a new setup from scratch. The pricing model for services is pay-per-use.

As an AWS customer, you can choose among different data centers. AWS data centers are distributed worldwide. For example, you can start a virtual machine in Japan in exactly the same way as you would start one in Ireland. This enables you to serve customers worldwide.

The map in figure 1.1 shows AWS's data centers. Access is limited to some of them: some data centers are accessible for U.S. government organizations only, and special conditions apply for the data centers in China. Additional data centers have been announced for Canada, Spain, Switzerland, Israel, UAE, India, Australia, and New Zealand.

Amazon Web Services is a global public cloud provider, and as such, it has to have a global network of infrastructure to run and manage its many growing cloud services that support customers around the world.



As of March 2024, AWS spans 105 Availability Zones within 33 geographic regions around the world, with announced plans for 18 more Availability Zones and 6 more AWS Regions in Malaysia, Mexico, New Zealand, the Kingdom of Saudi Arabia, Thailand, and the AWS European Sovereign Cloud.

The components that make up the AWS Global Infrastructure.

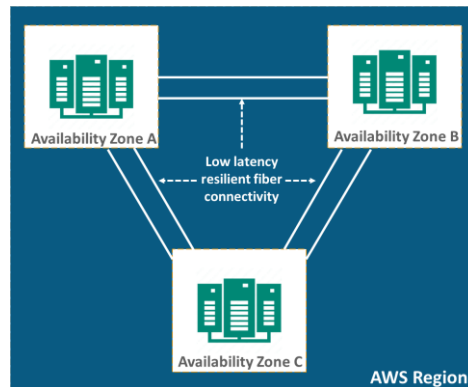
The components are:

- Availability Zones (AZs)
- Regions
- Edge Locations
- Regional Edge Caches
- Direct Connect Locations
- Point of Presence (POPs)
- Local Zones
- Wavelength Zones

If you are deploying services on AWS, you'll want to have a clear understanding of each of these components, how they are linked, and how you can use them within your solution to YOUR maximum benefit. Let's take a closer look.

Region:

- AWS has the notion of a Region, which is a physical location around the world where we cluster data centers. We call each group of logical data centers an Availability Zone.
- Cloud providers typically have multiple regions around the world, and each region is identified by a unique name (e.g., US East, Europe West, Asia Pacific).



Availability Zone:

- The term **Availability** refers to the ability of a system or component to be operational and accessible anytime.
- **Availability Zones** are locations carrying physical data centers in a **Region** which are used by customers, and each Region has **two or more AZs**
- An Availability Zone (AZ) is a data center or a group of data centers within a region that are isolated from each other but are connected through high-speed, low-latency networks.
- Each AZ has independent power supply, cooling system and Physical security.
- Each AZ is connected to other AZ in same Region via redundant and low latency network.
- AZ makes these applications and databases highly available, fault-tolerant, and scalable than traditional data centers.

Edge Locations:

Edge Locations are AWS sites deployed in major cities and highly populated areas across the globe. They far outnumber the number of availability zones available.

While Edge Locations are not used to deploy your main infrastructures such as EC2 instances, EBS storage, VPCs, or RDS resources like AZs, they are used by AWS services such as AWS

CloudFront and AWS Lambda@Edge (currently in Preview) to cache data and reduce latency for end-user access by using the Edge Locations as a global Content Delivery Network (CDN).

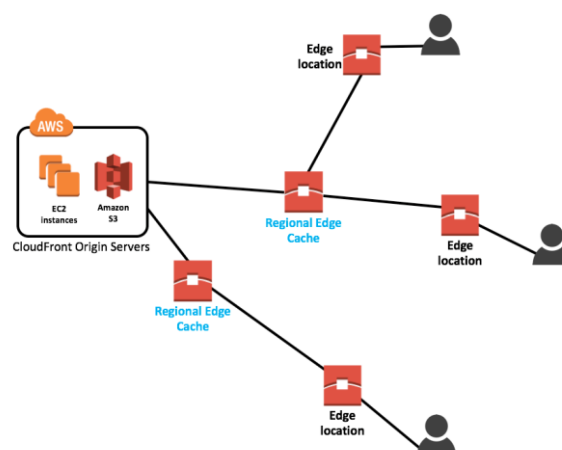
As a result, Edge Locations are primarily used by end users who are accessing and using your services.

For example, you may have your website hosted on EC2 instances and S3 (your origin) within the Ohio region with a configured CloudFront distribution associated. When a user accesses your website from Europe, they would be re-directed to their closest Edge Location (in Europe) where cached data could be read on your website, significantly reducing latency.

Regional Edge Cache:

In November 2016, AWS announced a new type of Edge Location, called a Regional Edge Cache. These sit between your CloudFront Origin servers and the Edge Locations. A Regional Edge Cache has a larger cache-width than each of the individual Edge Locations, and because data expires from the cache at the Edge Locations, the data is retained at the Regional Edge Caches.

Therefore, when data is requested at the Edge Location that is no longer available, the Edge Location can retrieve the cached data from the Regional Edge Cache instead of the Origin servers, which would have a higher latency.



Direct Connect Locations:

- These are the locations within the regions, where organizations can establish private connects between the AWS and their datacenter, office, or colocation environment.

- Such connections result reduced network costs, increased bandwidth throughput and more consistent network experience than Internet-based connections.

AWS Local Zones

- In 2022, Amazon announced that it had launched its first 16 Local Zones, a new type of infrastructure deployment designed to place core AWS Compute, Storage, Networking, and Database services near highly populated areas such as major cities that do not already have an AWS Region nearby.
- For example, the eastern United States has two Regions: us-east-1 in northern Virginia and us-east-2 in Ohio. However, there are also very large metropolitan areas around Boston, New York City, Philadelphia, Atlanta, and Miami, all of which are 100 miles or more from the data centers in that Region's nearest Availability Zones.
- AWS Local Zones allow customers in these areas to deploy resources and applications that require single-digit millisecond latency that would otherwise not be attainable given the geographic distance to the nearest Regions

AWS Wavelength Zones:

- Much like AWS Local Zones, AWS Wavelength Zones also place core AWS services closer to large end user bases and are connected to a parent Region via a secure, dedicated high-speed connection. However, AWS Wavelength Zones are embedded within 5G mobile broadband networks and are deployed within the data centers of large telecommunications providers.
- Deploying AWS resources such as VPC subnets, EC2 instances, and EBS volumes to an AWS Wavelength Zone allows end users to connect to these resources without ever leaving the mobile provider's network.

Point of Presence (PoP):

- A Point of Presence is not directly related to regions or data centers but is a network location with networking equipment, such as routers and switches. It's used to connect the cloud provider's infrastructure to the internet and other networks.
- **Points of Presence** are small setups in different locations where AWS caches the content and delivers it quickly to the end-user.
- PoPs are strategically located in various locations to improve network performance and reduce latency for end-users accessing cloud services.

Need of AWS

What can you do with AWS?

You can run all sorts of application on AWS by using one or a combination of services. The examples in this section will give you an idea of what you can do.

Hosting a web shop

John is CIO of a medium-sized e-commerce business. He wants to develop a fast and reliable web shop. He initially decided to host the web shop on-premises, and three years ago he rented machines in a data center. A web server handles requests from customers, and a database stores product information and orders. John is evaluating how his company can take advantage of AWS by running the same setup on AWS, as shown in fig.

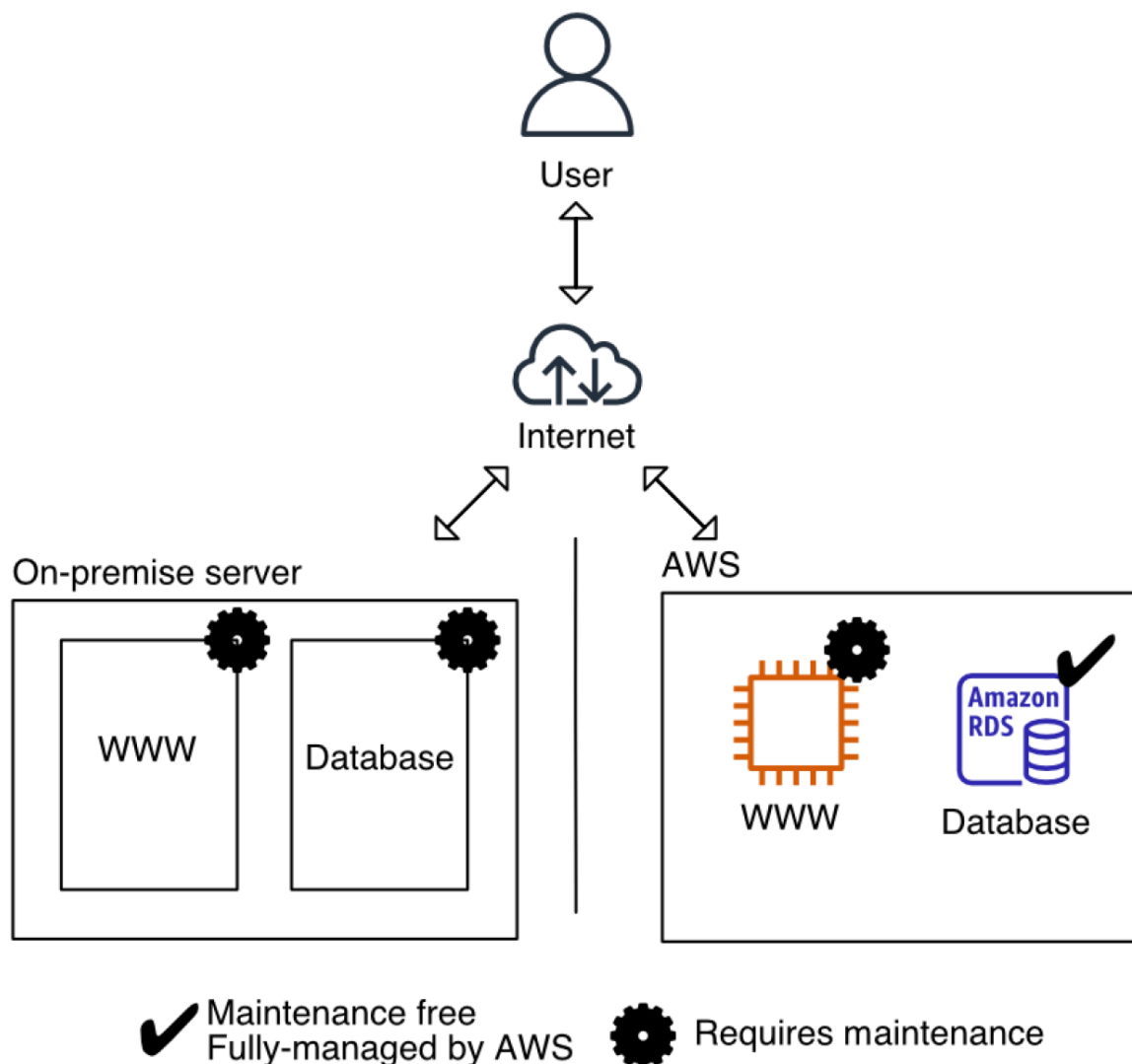


Fig. Running a web shop on-premises vs. on AWS

John not only wants to lift-and-shift his current on-premises infrastructure to AWS; he wants to get the most out of the advantages the cloud is offering. Additional AWS services allow John to improve his setup.

- The web shop consists of dynamic content (such as products and their prices) and static content (such as the company logo). Splitting these up would reduce the load on the web servers and improve performance by delivering the static content over a content delivery network (CDN).
- Switching to maintenance-free services including a database, an object store, and a DNS system would free John from having to manage these parts of the system, decreasing operational costs and improving quality.
- The application running the web shop can be installed on virtual machines. Using AWS, John can run the same amount of resources he was using on his on-premises machine, but split into multiple smaller virtual machines at no extra cost. If one of these virtual machines fails, the load balancer will send customer requests to the other virtual machines. This setup improves the web shop's reliability.

Fig. shows how John enhanced the web shop setup with AWS.

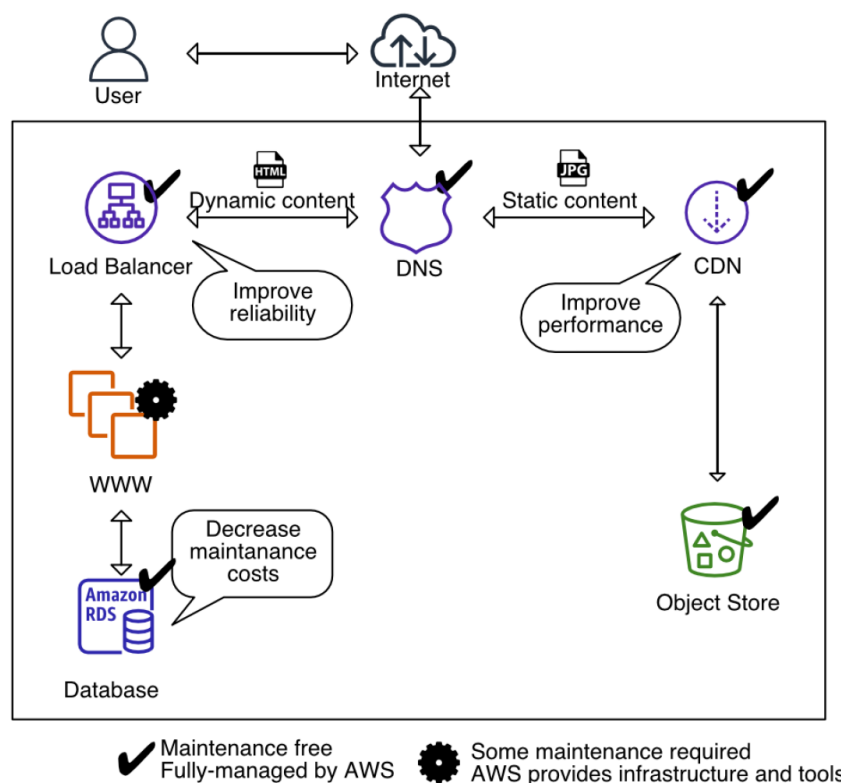


Figure 1.3 Running a web shop on AWS with CDN for better performance, a load balancer for high availability, and a managed database to decrease maintenance costs

John is happy with running his web shop on AWS. By migrating his company's infrastructure to the cloud, he was able to increase the reliability and performance of the web shop.

Running a Java EE application in your private network

Maureen is a senior system architect in a global corporation. She wants to move parts of her company's business applications to AWS when the data-center contract expires in a few months, to reduce costs and gain flexibility. She wants to run enterprise applications (such as Java EE applications) consisting of an application server and an SQL database on AWS. To do so, she defines a virtual network in the cloud and connects it to the corporate network through a Virtual Private Network (VPN) connection. She installs application servers on virtual machines to run the Java EE application. Maureen also wants to store data in an SQL database service (such as Oracle Database Enterprise Edition or Microsoft SQL Server EE).

For security, Maureen uses subnets to separate systems with different security levels from each other. By using access-control lists, she can control ingoing and outgoing traffic for each subnet. For example, the database is only accessible from the JEE server's subnet which helps to protect mission-critical data. Maureen controls traffic to the internet by using Network Address Translation (NAT) and firewall rules as well. Fig. illustrates Maureen's architecture.

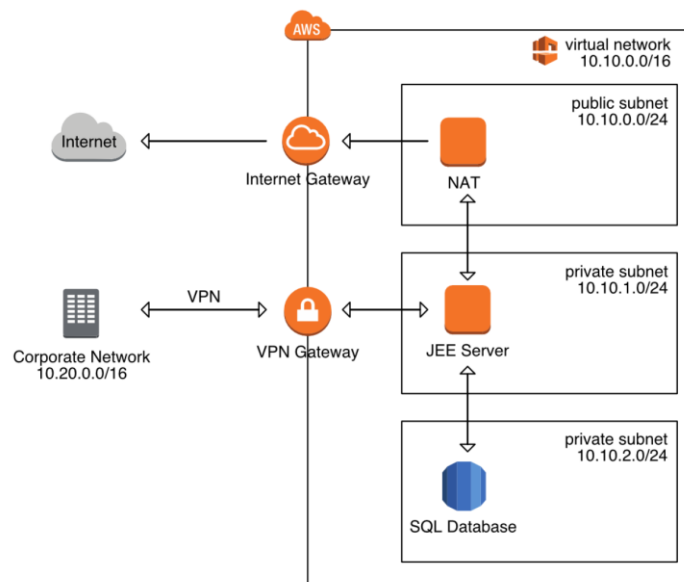


Figure 1.4 Running a Java EE application with enterprise networking on AWS improves flexibility and lowers costs.

Maureen has managed to connect the local data center with a private network running remotely on AWS to enable clients to access the JEE server. To get started, Maureen uses a VPN connection between the local data center and AWS, but she is already thinking about setting up

a dedicated network connection to reduce network costs and increase network throughput in the future.

The project was a great success for Maureen. She was able to reduce the time needed to set up an enterprise application from months to hours, as AWS can take care of the virtual machines, databases, and even the networking infrastructure on demand within a few minutes. Maureen's project also benefits from lower infrastructure costs on AWS, compared to using their own infrastructure on-premises.

Implementing a highly available system

Alexa is a software engineer working for a fast-growing startup. She knows that Murphy's Law applies to IT infrastructure: anything that can go wrong will go wrong. Alexa is working hard to build a highly available system to prevent outages from ruining the business. All services on AWS are either highly available or can be used in a highly available way. So, Alexa builds a system like the one shown in figure 1.5 with a high availability architecture. The database service is offered with replication and fail-over handling. In case the primary database instance fails, the standby database is promoted as the new primary database automatically. Alexa uses virtual machines acting as web servers. These virtual machines aren't highly available by default, but Alexa launches multiple virtual machines in different data centers to achieve high availability. A load balancer checks the health of the web servers and forwards requests to healthy machines.

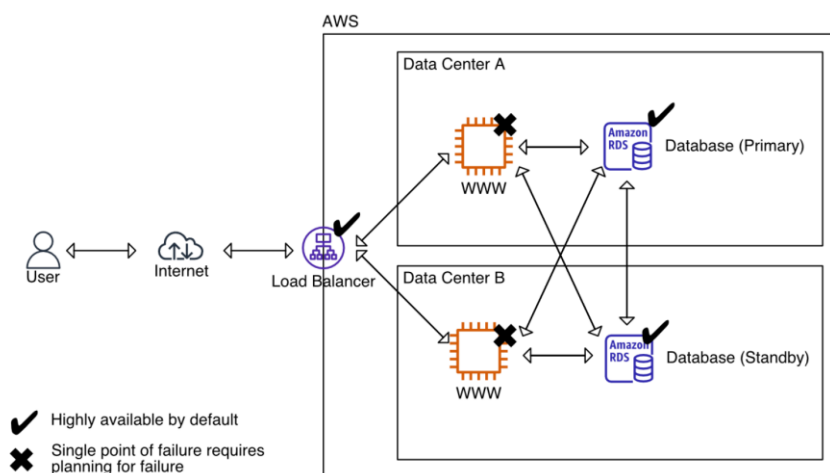


Figure 1.5 Building a highly available system on AWS by using a load balancer, multiple virtual machines, and a database with primary-standby replication

So far, Alexa has protected the startup from major outages. Nevertheless, she and her team are always planning for failure and are constantly improving the resilience of their systems.

Profiting from low costs for batch processing infrastructure

Nick is a data scientist who needs to process massive amounts of measurement data collected from gas turbines. He needs to generate a report containing the maintenance condition of hundreds of turbines daily. Therefore, his team needs a computing infrastructure to analyze the newly arrived data once a day. Batch jobs are run on a schedule and store aggregated results in a database. A business intelligence (BI) tool is used to generate reports based on the data stored in the database.

As the budget for computing infrastructure is very small, Nick and his team have been looking for a cost effective solution to analyze their data. He finds a way to make clever use of AWS's price model:

- AWS bills virtual machines per second with a minimum of 60 seconds and. So Nick launches a virtual machine when starting a batch job, and terminates it immediately after the job finished. Doing so allows him to pay for computing infrastructure only when actually using it. This is a big game changer compared to the traditional data center where Nick had to pay a monthly fee for each machine, no matter how much it was used.
- AWS offers spare capacity in their data centers at substantial discount. It is not important for Nick to run a batch job at a specific time. He can wait to execute a batch job until there is enough spare capacity available, so AWS offers him a virtual machine with a discount of 50%.

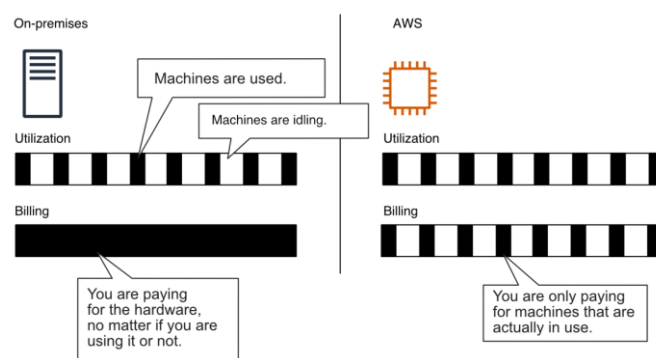


Figure 1.6 Making use of the pay-per-use price model of virtual machines

Nick is happy to have access to a computing infrastructure that allows his team to analyze data at low costs. You now have a broad idea of what you can do with AWS. Generally speaking, you can host any application on AWS. The next section explains the nine most important benefits AWS has to offer.

Benefits of AWS

Using Amazon Web Services (AWS) offers numerous benefits that contribute to the success and efficiency of businesses. Here's a more detailed breakdown of the points you mentioned:

Innovative and Fast-Growing Platform:

- AWS constantly evolves, introducing new services and features to stay at the forefront of cloud technology.
- Access to cutting-edge technologies, such as machine learning, AI, IoT, and serverless computing.

Services Solve Common Problems:

- A wide range of services tailored to address common business challenges, from computing and storage to databases, analytics, and security.

Enabling Automation:

- AWS provides tools and services for automating manual and repetitive tasks, allowing businesses to increase efficiency and reduce operational overhead.

Flexible Capacity (Scalability):

- Easily scale resources up or down based on demand, ensuring optimal performance and cost efficiency.
- Pay-as-you-go pricing model allows businesses to only pay for the resources they use.

Built for Failure (Reliability):

- AWS infrastructure is designed to be highly resilient and fault-tolerant.
- Multi-region and multi-availability zone options reduce the risk of downtime due to hardware failures or other issues.

Reducing Time to Market:

- Rapid deployment of infrastructure and applications allows businesses to bring products and services to market faster.
- Pre-built templates and managed services accelerate development cycles.

Benefiting from Economies of Scale:

- Shared infrastructure and resources across a vast customer base result in cost savings for individual businesses.
- Large-scale operations enable AWS to negotiate better deals with hardware and software providers.

Global Infrastructure:

- Presence in multiple geographic regions ensures low-latency access for users worldwide.
- The ability to deploy resources in different regions enhances data redundancy and compliance with regional regulations.

Professional Partner:

- Access to a vast ecosystem of AWS partners and certified professionals.
- Expert support and guidance for planning, implementing, and optimizing AWS solutions.

By leveraging these aspects of AWS, businesses can achieve greater agility, cost savings, and innovation, allowing them to focus more on their core competencies while AWS manages the underlying infrastructure and services.



Exploring AWS services

In this section you will get an idea of the range of services that AWS offers. We'll also give you a mental model to help you better understand AWS.

Hardware for computing, storing, and networking is the foundation of the AWS cloud. AWS runs services on this hardware, as shown in figure 1.9. The API acts as an interface between AWS services and your applications.

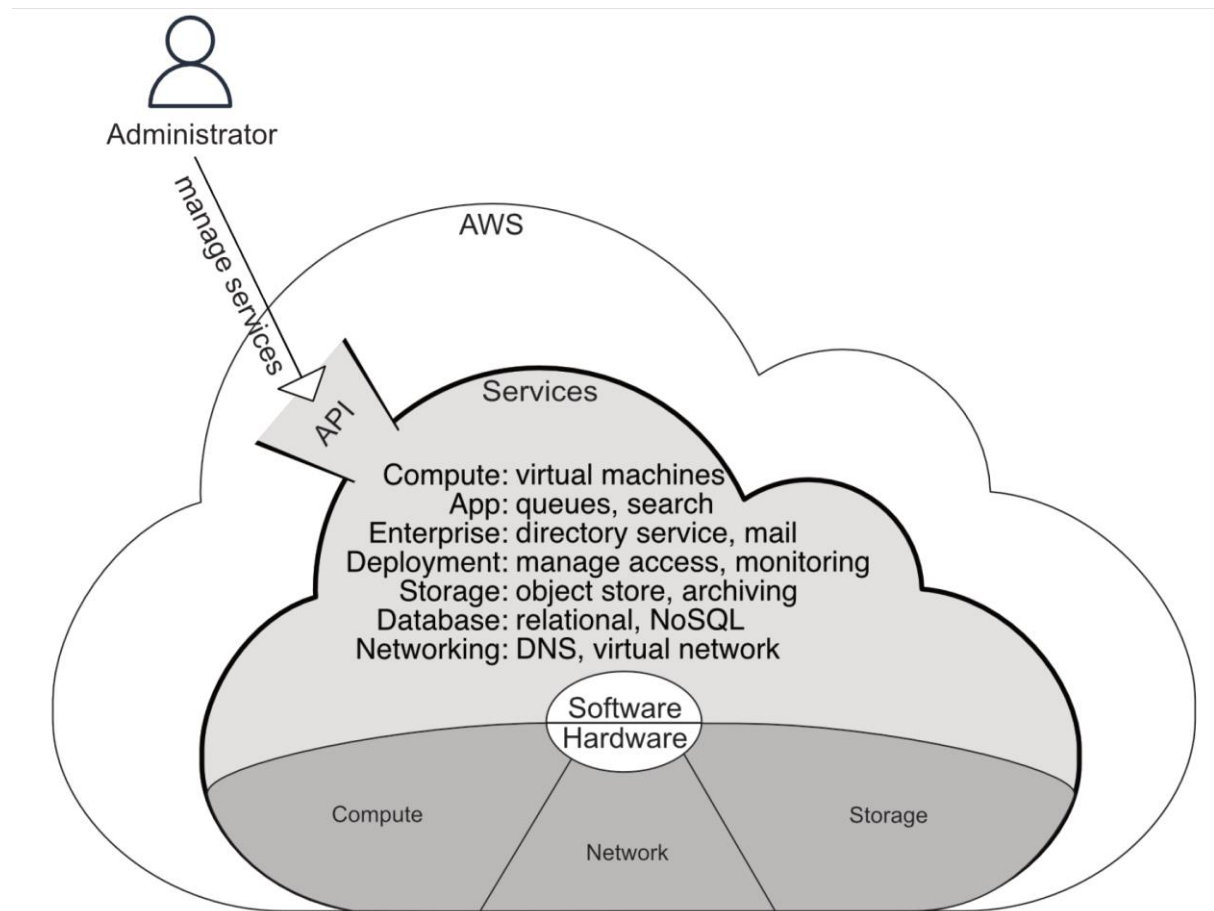


Figure 1.9 The AWS cloud is composed of hardware and software services accessible via an API.

You can manage services by sending requests to the API manually via a web-based GUI like the Management Console, a command-line interface (CLI), or programmatically via an SDK. Virtual machines have a special feature: you can connect to virtual machines through SSH, for example, and gain administrator access. This means you can install any software you like on a virtual machine. Other services, like the NoSQL database service, offer their features through an API and hide everything that's going on behind the scenes. Figure 1.10 shows an

administrator installing a custom PHP web application on a virtual machine and managing dependent services such as a NoSQL database used by the application.

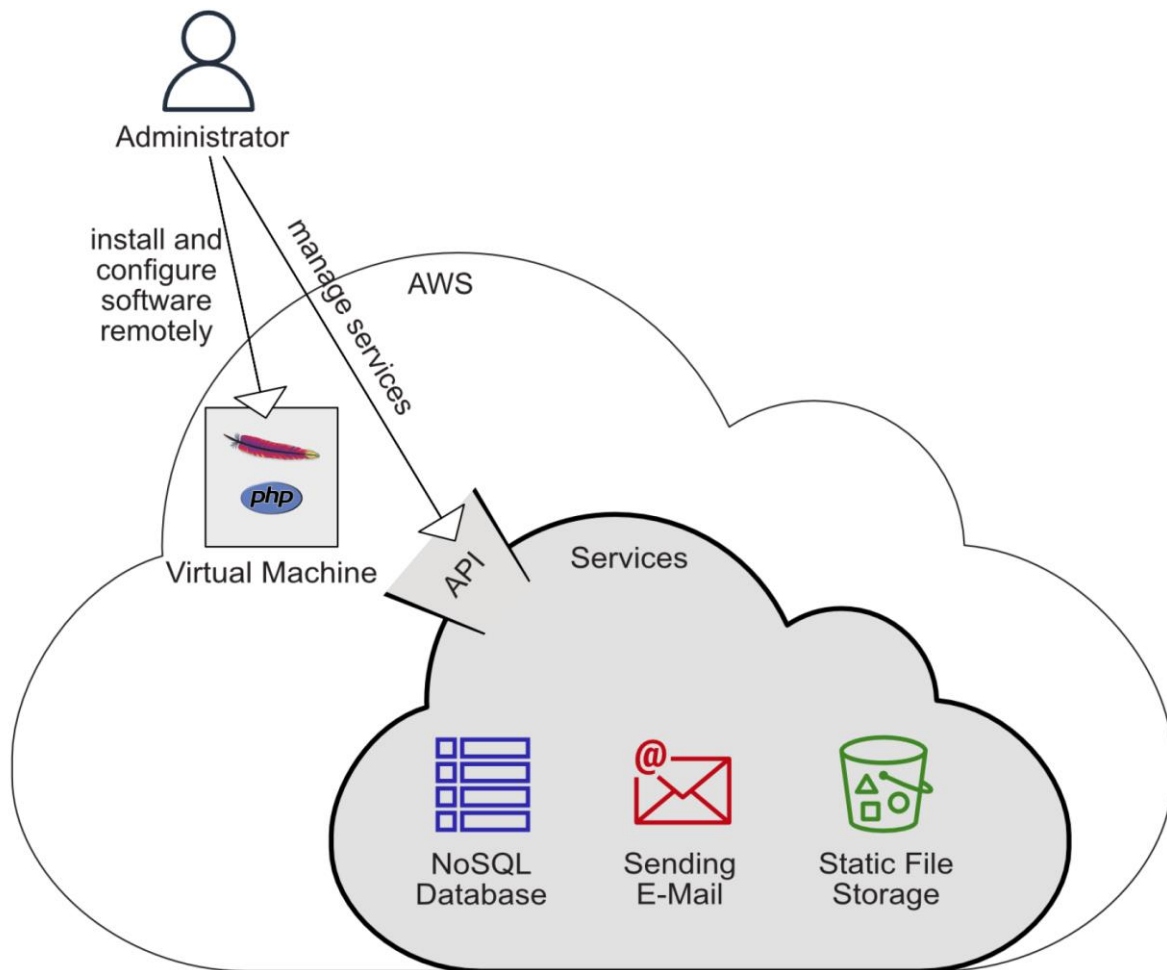


Figure 1.10 Managing a custom application running on a virtual machine and cloud-native services

Users send HTTP requests to a virtual machine. This virtual machine is running a web server along with a custom PHP web application. The web application needs to talk to AWS services in order to answer HTTP requests from users. For example, the application might need to query data from a NoSQL database, store static files, and send email. Communication between the web application and AWS services is handled by the API, as figure 1.11 shows.

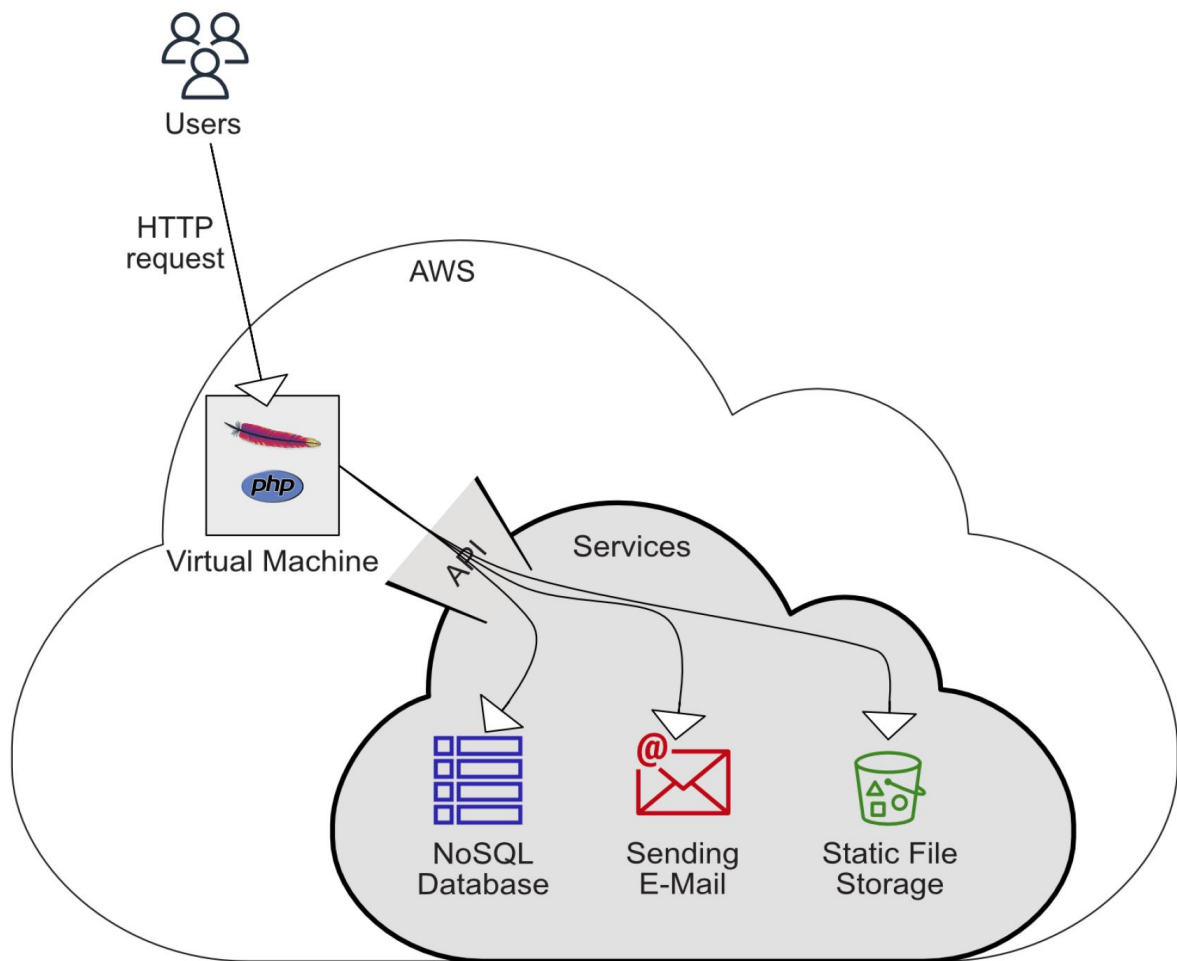


Figure 1.11 Handling an HTTP request with a custom web application using additional AWS services

The number of services available can be scary at the outset. When logging into AWS's web interface you are presented with an overview listing around 200 services. On top of that, new services are announced constantly during the year and at the big conference in Las Vegas, AWS re:Invent.

AWS offers services in the following categories:

• Analytics	• Application Integration	• AR & VR
• AWS Cost Management	• Blockchain	• Business Applications
• Compute	• Containers	• Customer Enablement
• Database	• Developer Tools	• End User Computing
• Front-end Web & Mobile	• Game Development	• Internet of Things
• Machine Learning	• Management & Governance	• Media Services
• Migration & Transfer	• Networking & Content Delivery	• Quantum Technologies
• Robotics	• Satellite	• Security, Identity, & Compliance
• Storage		

It is impossible to cover the many services offered by AWS in one book. Therefore, we are focusing on the services that will best help you get started quickly, as well as the most widely used services. The following services are covered in detail in our book:

- EC2—Virtual machines
- ECS and Fargate—Running and managing containers
- Lambda—Executing functions
- S3—Object store
- Glacier—Archiving data
- EBS—Block storage for virtual machines
- EFS—Network filesystem
- RDS—SQL databases

- DynamoDB—NoSQL database
- ElastiCache—In-memory key-value store
- VPC—Virtual network
- ELB—Load balancers
- Simple Queue Service—Distributed queues
- CodeDeploy—Automate code deployments
- CloudWatch—Monitoring and logging
- CloudFormation—Automating your infrastructure
- IAM—Restricting access to your cloud resources

We are missing at least three important topics that would fill their own books: continuous delivery, machine learning, and analytics. "Let us know when you are interested in reading one of these unwritten books."

But how do you interact with an AWS service? The next section explains how to use the web interface, the CLI, and SDKs to manage and access AWS resources.

Interacting with AWS

When you interact with AWS to configure or use services, you make calls to the API. The API is the entry point to AWS, as figure 1.12 demonstrates.

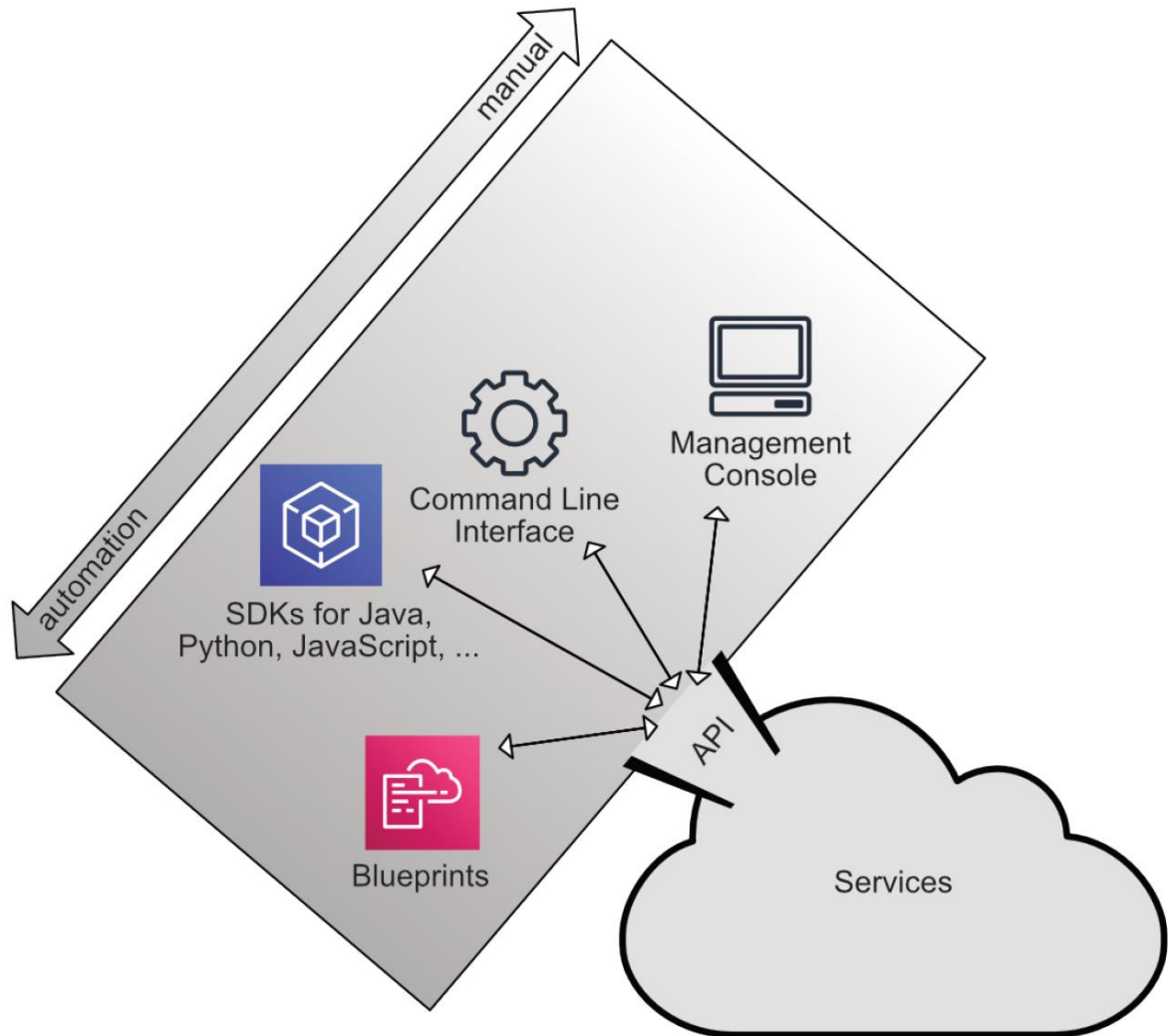


Figure 1.12 Different ways to access the AWS API, allowing you to manage and access AWS services

Next, we'll give you an overview of the tools available for communicating with AWS's APIs: the Management Console, the command-line interface, the SDKs, and infrastructure blueprints. We will compare the different tools, and you will learn how to use all of them while working your way through the book.

The AWS Management Console allows you to manage and access AWS services through a

graphical user interface (GUI), which works with modern browsers on desktop computers, laptops, and tablets. See figure 1.13.

1.7.1 Management Console

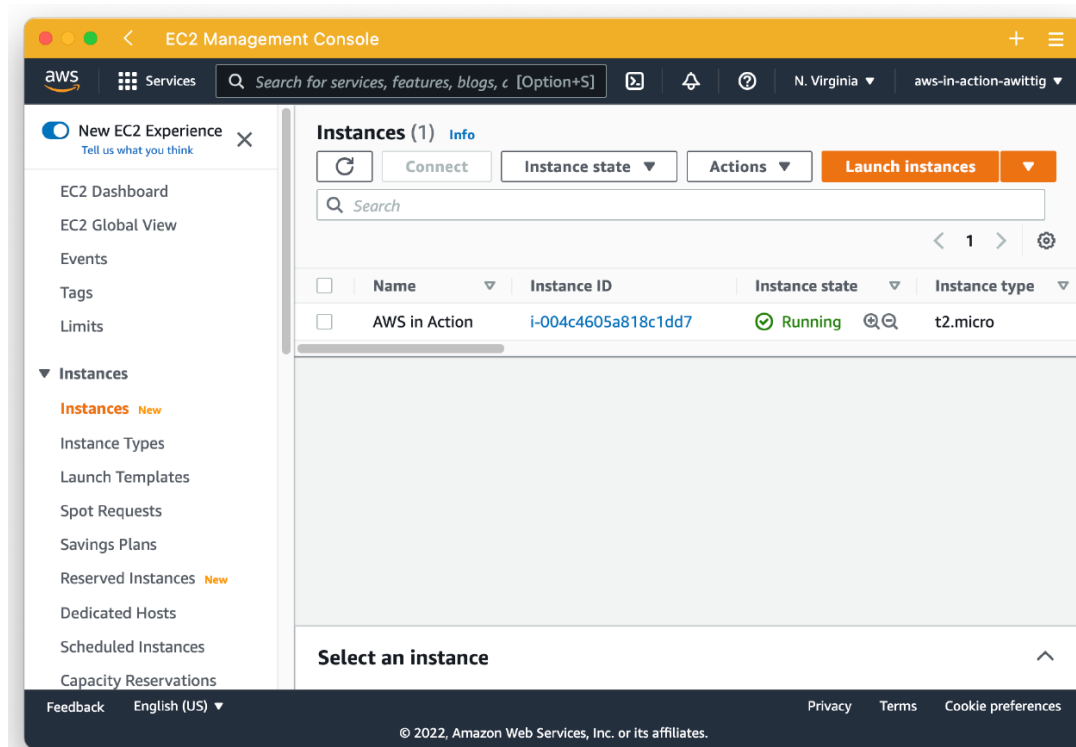


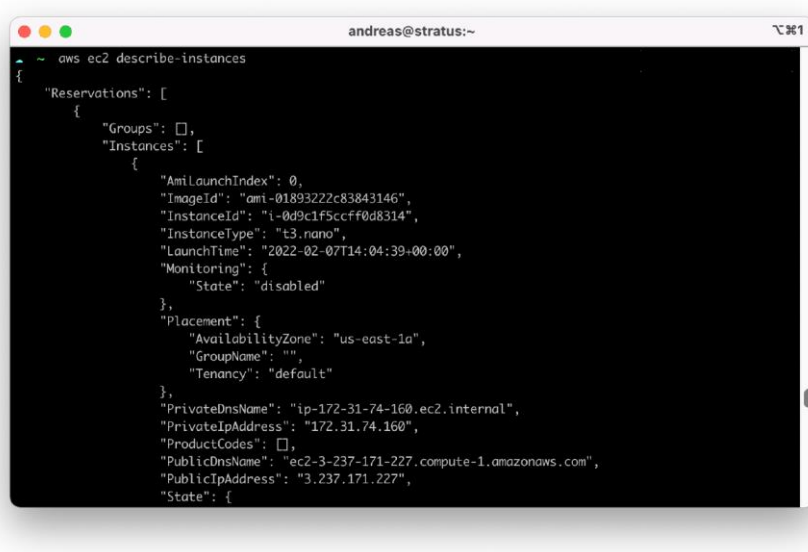
Figure 1.13 The AWS Management Console offers a GUI to manage and access AWS services.

When getting started or experimenting with AWS, the Management Console is the best place to start. It helps you to gain an overview of the different services quickly. The Management Console is also a good way to set up a cloud infrastructure for development and testing.

- Web-based graphical interface for managing AWS resources
- Best for beginners and manual operations
- Access at: <https://console.aws.amazon.com>
- Features:
 - Point-and-click interface
 - Visual representation of resources
 - Built-in wizards for common tasks

1.7.2 Command-line interface

The command-line interface (CLI) allows you to manage and access AWS services within your terminal. Because you can use your terminal to automate or semi-automate recurring tasks, CLI is a valuable tool. You can use the terminal to create new cloud infrastructures based on blueprints, upload files to the object store, or get the details of your infrastructure's networking configuration regularly. Figure 1.14 shows the CLI in action.



```
andreas@stratus:~  
~ aws ec2 describe-instances  
{  
  "Reservations": [  
    {  
      "Groups": [],  
      "Instances": [  
        {  
          "AmiLaunchIndex": 0,  
          "ImageId": "ami-01893222c83843146",  
          "InstanceId": "i-0d9c1f5ccff0d8314",  
          "InstanceType": "t3.nano",  
          "LaunchTime": "2022-02-07T14:04:39+00:00",  
          "Monitoring": {  
            "State": "disabled"  
          },  
          "Placement": {  
            "AvailabilityZone": "us-east-1a",  
            "GroupName": "",  
            "Tenancy": "default"  
          },  
          "PrivateDnsName": "ip-172-31-74-160.ec2.internal",  
          "PrivateIpAddress": "172.31.74.160",  
          "ProductCodes": [],  
          "PublicDnsName": "ec2-3-237-171-227.compute-1.amazonaws.com",  
          "PublicIpAddress": "3.237.171.227",  
          "State": {
```

Figure 1.14 The CLI allows you to manage and access AWS services from your terminal.

If you want to automate parts of your infrastructure with the help of a continuous integration server, like Jenkins, the CLI is the right tool for the job. The CLI offers a convenient way to access the API and combine multiple calls into a script.

You can even begin to automate your infrastructure with scripts by chaining multiple CLI calls together. The CLI is available for Windows, Mac, and Linux, and there is also a PowerShell version available.

Configuring AWS CLI

Before working with S3, you need to configure your AWS CLI with your credentials:

aws configure

You'll be prompted for:

AWS Access Key ID

AWS Secret Access Key

Default region name (e.g., us-east-1)

Default output format (e.g., json, text, or table)

Basic S3 Operations

List Buckets: `aws s3 ls`

Create a Bucket: `aws s3 mb s3://bucket-name`

Remove a Bucket: `aws s3 rb s3://bucket-name`

1.7.3 SDKs

Use your favorite programming language to interact with the AWS API. AWS offers SDKs for the following platforms and languages:

• JavaScript	• Python	• PHP
• .NET	• Ruby	• Java
• Go	• Node.js	• C++

SDKs are typically used to integrate AWS services into applications. If you're doing software development and want to integrate an AWS service like a NoSQL database or a push-notification service, an SDK is the right choice for the job. Some services, such as queues and topics, must be used with an SDK.

1.7.4 Blueprints

A blueprint is a description of your system containing all resources and their dependencies. An Infrastructure as Code tool compares your blueprint with the current system, and calculates the steps to create, update, or delete your cloud infrastructure. Figure 1.15 shows how a blueprint is transferred into a running system.

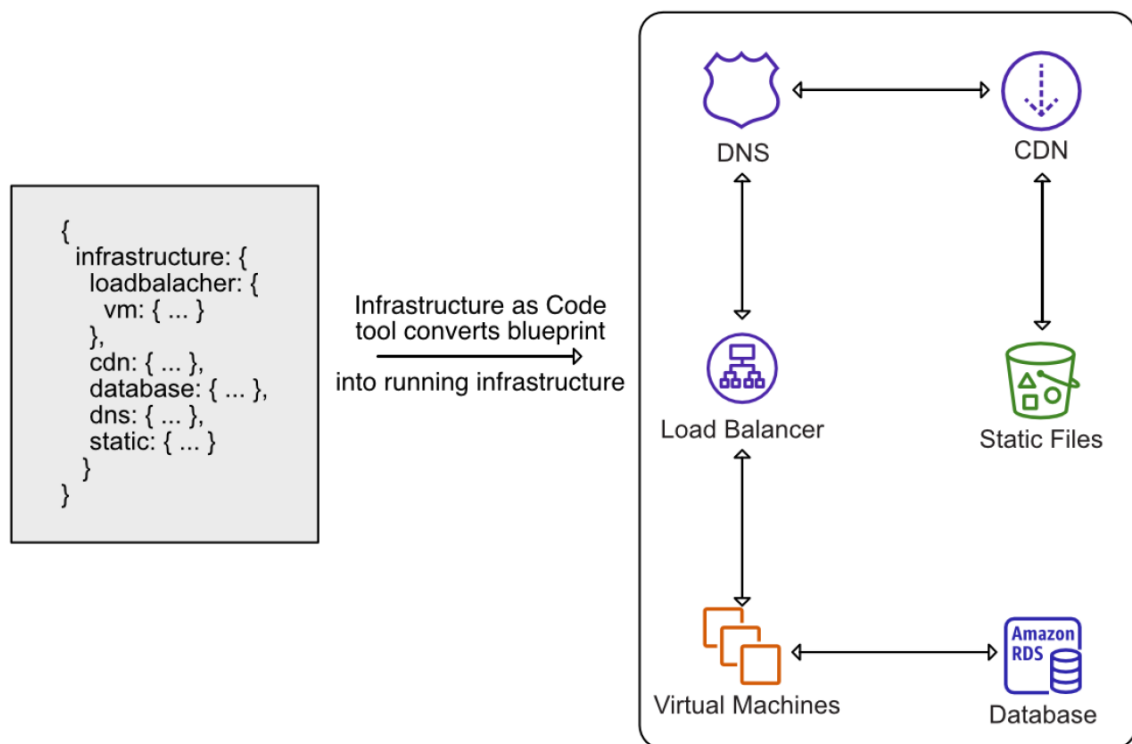


Figure 1.15 Infrastructure automation with blueprints

Consider using blueprints if you have to control many or complex environments. Blueprints will help you to automate the configuration of your infrastructure in the cloud. You can use them to set up a network and launch virtual machines, for example.

Automating your infrastructure is also possible by writing your own source code with the help of the CLI or the SDKs. But doing so requires you to resolve dependencies, make sure you are able to update different versions of your infrastructure, and handle errors yourself.

Creating an AWS account

Signing up for a Free Tier Account with AWS

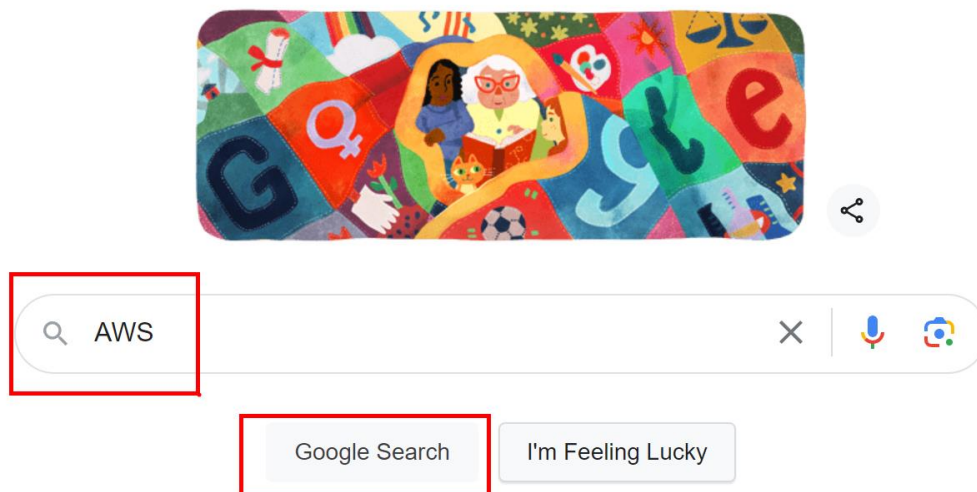
Before you can start using AWS, you need to create an account. Your account is a basket for all your cloud resources. You can attach multiple users to an account if multiple humans need access to it; by default, your account will have one AWS account root user. To create an account, you need the following:

- A telephone number to validate your identity
- A credit card to pay your bills

The sign-up process consists of five steps:

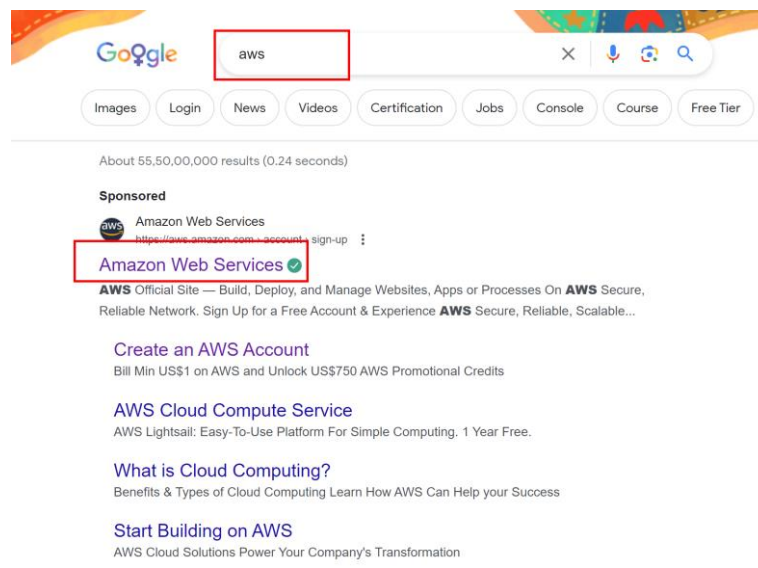
- 1) Providing login credentials
- 2) Providing contact information
- 3) Providing payment details
- 4) Verifying your identity
- 5) Choosing a support plan

- 1) Open a browser and Type “AWS” click on “Google Search” button

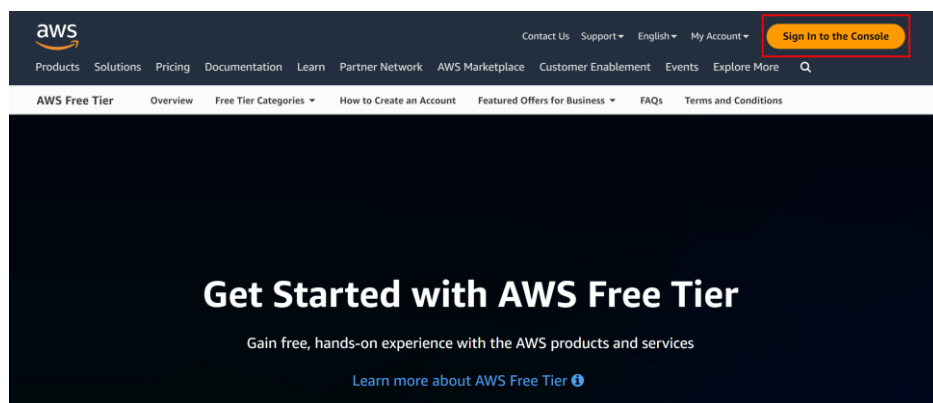


Google offered in: [हिन्दी](#) [বাংলা](#) [తెలుగు](#) [मराठी](#) [தமிழ்](#) [ગુજરાતી](#) [ಕನ್ನಡ](#) [മലയാളം](#) [ਪੰਜਾਬੀ](#)

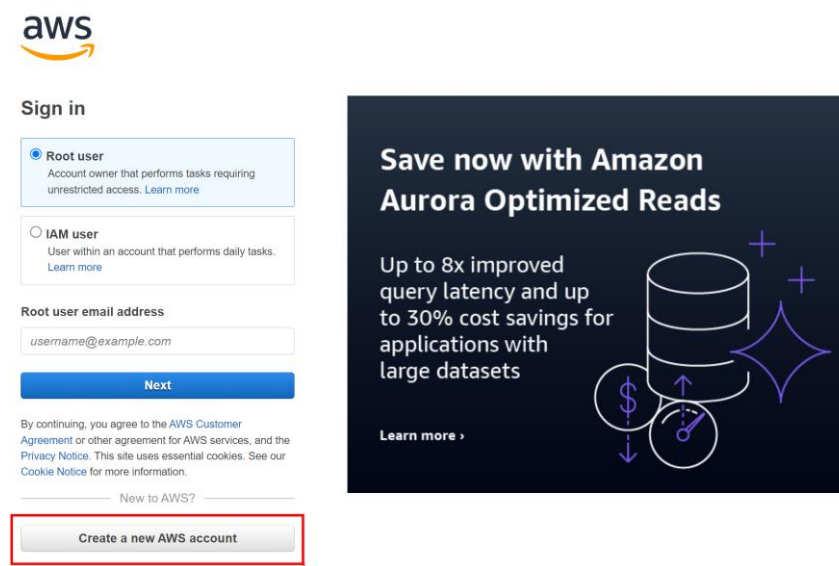
2) Click on the First link “Amazon Web Services”



3) Click on the “Sign in to the Console”



4) Click on the “Create a new AWS account”



5) Enter the Root user email address and AWS account name in the Text Boxes

Creating an AWS account starts with defining a unique AWS account name. The AWS account name has to be globally unique among all AWS customers.



Explore Free Tier products with a new AWS account.

To learn more, visit aws.amazon.com/free.



Sign up for AWS

Root user email address
Used for account recovery and some administrative functions

AWS account name
Choose a name for your account. You can change this name in your account settings after you sign up.

Verify email address

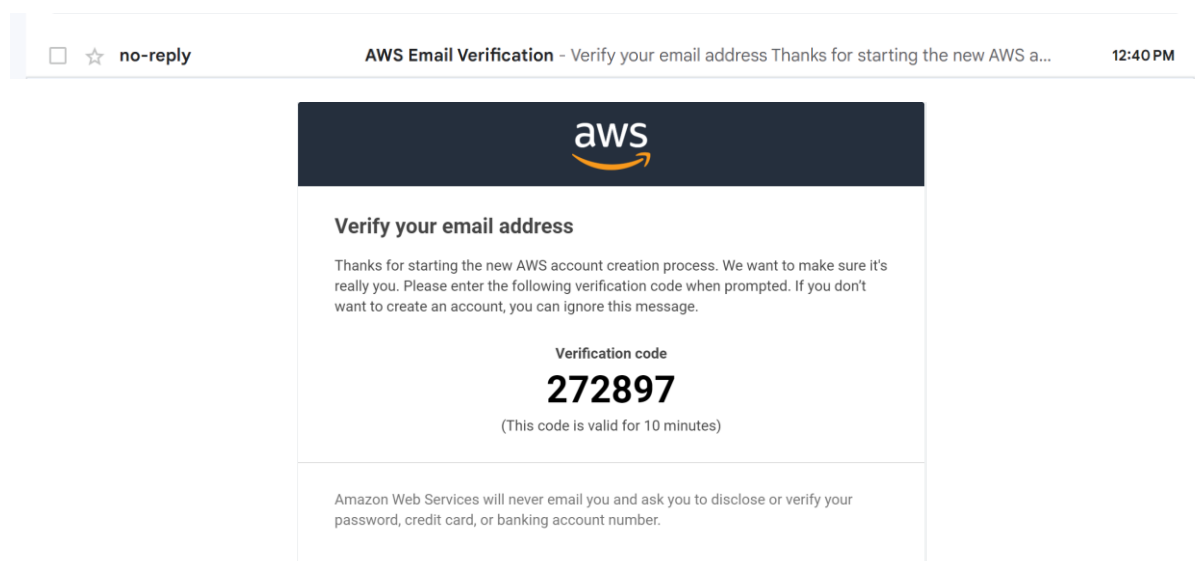
OR

Sign in to an existing AWS account

6) Redirect to the Email provided as Root user email address.

You will receive a Verification mail.

Open the mail and copy the code provisioned in the mail.



7) Paste the Verification code and click on the “Verify”



Explore Free Tier products with a new AWS account.

To learn more, visit aws.amazon.com/free.



Explore Free Tier products with a new AWS account.

To learn more, visit aws.amazon.com/free.



Sign up for AWS

Confirm you are you

Making sure you are secure -- it's what we do.

We sent an email with a verification code to **krr29101991@gmail.com**. ([not you?](#))

Enter it below to confirm your email.

Verification code

Verify

Resend code

Didn't get the code?

- Codes can take up to 5 minutes to arrive.
- Check your spam folder.

Sign up for AWS

Confirm you are you

Making sure you are secure -- it's what we do.

We sent an email with a verification code to **krr29101991@gmail.com**. ([not you?](#))

Enter it below to confirm your email.

Verification code

272897

Verify

Resend code

Didn't get the code?

- Codes can take up to 5 minutes to arrive.
- Check your spam folder.

8) Enter your choice of Password and click on Continue

Explore Free Tier products with a new AWS account.

To learn more, visit aws.amazon.com/free.



Explore Free Tier products with a new AWS account.

To learn more, visit aws.amazon.com/free.



Sign up for AWS

Create your password

✓ It's you! Your email address has been successfully verified. ✕

Your password provides you with sign in access to AWS, so it's important we get it right.

Root user password

Confirm root user password

Continue (step 1 of 5)

OR

Sign in to an existing AWS account

Sign up for AWS

Create your password

✓ It's you! Your email address has been successfully verified. ✕

Your password provides you with sign in access to AWS, so it's important we get it right.

Root user password

Confirm root user password

Continue (step 1 of 5)

OR

Sign in to an existing AWS account

9) Fill the Details



Free Tier offers

All AWS accounts can explore 3 different types of free offers, depending on the product used.



Always free
Never expires



12 months free
Start from initial sign-up date



Trials
Start from service activation date

Sign up for AWS

Contact Information

How do you plan to use AWS?

- ☐ Business - for your work, school, or organization
- ☐ Personal - for your own projects

Who should we contact about this account?

Full Name

Phone Number

 +1 ▼

222-333-4444

Country or Region

United States ▼

Address

City

State, Province, or Region

Postal Code

☐ I have read and agree to the terms of the [AWS Customer Agreement](#).

Continue (step 2 of 5)



Free Tier offers

All AWS accounts can explore 3 different types of free offers, depending on the product used.



Always free
Never expires



12 months free
Start from initial sign-up date



Trials
Start from service activation date

Sign up for AWS

Contact Information

How do you plan to use AWS?

- ☐ Business - for your work, school, or organization
- ☒ Personal - for your own projects

Who should we contact about this account?

Full Name

K Vishal Reddy

Phone Number

+91 9405912849

Country or Region

India

Address

HNo.2-68/51/85, Street No.8 Left,

Apartment, suite, unit, building, floor, etc.

City

Hyderabad

State, Province, or Region

TELANGANA

Postal Code

500092

Customers with an Indian contact address are served by Amazon Web Services India Private Limited, the local seller for AWS services in India.

☒ I have read and agree to the terms of the [AWS Customer Agreement](#).

Continue (step 2 of 5)

10) Fill the Credit card Details



Secure verification

 We will not charge you for usage below AWS Free Tier limits. We may temporarily hold up to \$1 USD (or an equivalent amount in local currency) as a pending transaction for 3-5 days to verify your identity.



Sign up for AWS

Billing Information

Credit or Debit card number

 The credit card number is required.



AWS accepts most major credit and debit cards. To learn more about payment options, review our [FAQ](#)

Expiration date

June

▼

2029

▼

Security code 

Cardholder's name

 The cardholder's name is required.

☐ Save card information for faster future payments
Securely save card information payments as per RBI guidelines. [Learn more.](#)

Billing address

☒ Use my contact address
HNo.2-68/51/85, Street No.8 Left,
Hyderabad TELANGANA 500092
IN

☐ Use a new address

Do you have a PAN?

Permanent Account Number (PAN) is a ten-digit alphanumeric number issued by the Indian Income Tax Department. This 10-digit number is printed on the front of your PAN card.

☐ Yes

☐ No

You can go on the Tax Settings Page on Billing and Cost Management Console to update your PAN information.

Verify and Continue (step 3 of 5)

You might be redirected to your bank's website to authorize the verification charge.

11) Confirm your identity



Sign up for AWS

Confirm your identity [Info](#)

Name [Info](#)

Choose the name that you want to use for identity verification.

☐ K Vishal Reddy

Primary purpose of account registration

Choose one that best applies to you. If your account is tied to a business, select the one that applies to your business.

Select one ▼

Ownership type

Select one ▼

☐ I consent to allowing AWS to use and send the information above to a third-party service for identity verification purposes.

Continue (step 4 of 5)



Sign up for AWS

Confirm your identity [Info](#)

Name [Info](#)

Choose the name that you want to use for identity verification.

☒ K Vishal Reddy

Primary purpose of account registration

Choose one that best applies to you. If your account is tied to a business, select the one that applies to your business.

Personal use ▼

Ownership type

Individual ▼

India document type [Info](#)

To verify your identity, the name on the document must match the name that you chose.

Permanent Account Number (PAN) ▼

Permanent Account Number (PAN)

CGSPK7132B

The PAN is 10 alphanumeric characters without spaces or tabs. Example: AAAAAA1111B

☒ I consent to allowing AWS to use and send the information above to a third-party service for identity verification purposes.

Continue (step 4 of 5)



Sign up for AWS

Confirm your identity

Before you can use your AWS account, you must verify your phone number. When you continue, the AWS automated system will contact you with a verification code.

How should we send you the verification code?

- ☒ Text message (SMS)
☐ Voice call

Country or region code

India (+91)

Mobile phone number

9405912849

Security check



Type the characters as shown above

z6xm2x

Send SMS (step 4 of 5)



Sign up for AWS

Confirm your identity

Verify code

Continue (step 4 of 5)

Having trouble? Sometimes it takes up to 10 minutes to retrieve a verification code. If it's been longer than that, [return to the previous page](#) and try again.

12) Choosing a support plan

For now, select the Basic plan, which is free. When running a production workload on AWS, we recommend at least a Developer plan to be able to ask questions about upcoming issues.



Sign up for AWS

Select a support plan

Choose a support plan for your business or personal account. [Compare plans and pricing examples](#)
[You can change your plan anytime in the AWS Management Console.](#)

☒ Basic support - Free - Recommended for new users just getting started with AWS 24x7 self-service access to AWS resources - For account and billing issues only - Access to Personal Health Dashboard & Trusted Advisor 	☐ Developer support - From \$29/month - Recommended for developers experimenting with AWS - Email access to AWS Support during business hours 12 (business)-hour response times 	☐ Business support - From \$100/month - Recommended for running production workloads on AWS 24x7 tech support via email, phone, and chat 1-hour response times - Full set of Trusted Advisor best-practice recommendations 
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Need Enterprise level support?

From \$15,000 a month you will receive 15-minute response times and concierge-style experience with an assigned Technical Account Manager. [Learn more](#)

[Complete sign up](#)



Congratulations

Thank you for signing up for AWS.

We are activating your account, which should only take a few minutes. You will receive an email when this is complete.

[Go to the AWS Management Console](#)

[Sign up for another account or contact sales.](#)

Signing In

You now have an AWS account and are ready to sign in to the AWS Management Console. As mentioned earlier, the Management Console is a web-based tool you can use to control AWS resources; it makes most of the functionality of the AWS API available to you. Choose *Root user* and enter your email address, click Next, and then enter your password to sign in.



Sign in

☒ **Root user**
Account owner that performs tasks requiring unrestricted access. [Learn more](#)

☐ **IAM user**
User within an account that performs daily tasks. [Learn more](#)

Root user email address

Next

By continuing, you agree to the [AWS Customer Agreement](#) or other agreement for AWS services, and the [Privacy Notice](#). This site uses essential cookies. See our [Cookie Notice](#) for more information.

☐ New to AWS?

Create a new AWS account

Save now with Amazon Aurora Optimized Reads

Up to 8x improved query latency and up to 30% cost savings for applications with large datasets

[Learn more >](#)

After you have signed in successfully, you are forwarded to the start page of the Management Console



Congratulations

Thank you for updating your Amazon Web Services (AWS) subscription. Your management console will now be up to date with all AWS services.

Go to the AWS Management Console

The most important part is the navigation bar at the top, shown in figure 1.24. It consists of seven sections:

1. *AWS*—The dashboard of the Management Console shows an overview of your AWS account.
2. *Services*—Provides access to all AWS services.
3. *Search*—Allows you to search for services, features, and more.
4. *Terminal*—Spin up a terminal with access to your cloud resources in the browser.
5. *Notifications*—View alerts and notifications by AWS, for example planned downtimes or outages.
6. *Help*—Get support by the community, experts, or AWS support.
7. *Region*—Select the region you want to manage.
8. *Account*—Manage your AWS account.

